

Ph.D. Dissertation of Science

Writing a Quarto-Based Thesis for Reproducible Data Analysis

Integrating R Analysis and Document Formatting

재현 가능한 데이터 분석을 위한 Quarto 기반 학위논문 작성

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Abstract

This document is a Quarto starter example for writing a thesis at Seoul National University. It demonstrates how thesis metadata, manuscript chapters, and R analysis can be maintained separately and rendered into one PDF. Students enter their thesis information in a configuration file and write each chapter in a QMD file.

The example uses the cars dataset distributed with R to examine the association between speed and stopping distance. Tables, figures, equations, references, and an appendix demonstrate the supported document components. These results illustrate the template and do not represent an actual thesis. Replace this abstract with the objectives, methods, results, and conclusions of your own research before submission.

Keywords: reproducibility, data analysis, Quarto, R, thesis

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Chapter 1

Introduction

1.1 Background

Manually copying analysis results into a manuscript can create inconsistencies between the code and the text. R provides a statistical computing environment (R Core Team 2026), and Quarto integrates executable analysis with document authoring (Posit n.d.).

This unofficial starter follows the document sequence provided by the SNU Library. Department-specific requirements must be checked before submission.

1.2 Objectives

The example demonstrates how to update a manuscript and its analysis together. Chapter 2 describes the methods, and Chapter 3 presents the results.

A footnote can contain supplementary information.¹

¹This is an example footnote.

Chapter 2

Methods

2.1 Data

The analysis uses the cars dataset distributed with R. Speed is measured in miles per hour, and stopping distance is measured in feet. Each chapter reads the shared analysis script into its own R session.

2.2 Statistical model

The relationship between speed and stopping distance is represented by a simple linear regression:

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \epsilon_i \quad (2.1)$$

The coefficients of Equation 2.1 are estimated using ordinary least squares.

Chapter 3

Results

3.1 Descriptive statistics

The dataset contains 50 observations. Mean speed is 15.4 mph, and mean stopping distance is 42.98 ft.

Table 3.1: Descriptive statistics for speed and stopping distance

Variable	Mean	SD
Speed (mph)	15.40	5.29
Stopping distance (ft)	42.98	25.77

Table 3.1 presents the mean and standard deviation of both variables.

3.2 Speed and stopping distance

Figure 3.1 shows that stopping distance tends to increase with speed. The estimated slope is 3.93. These results illustrate the template and are not findings from an actual thesis.

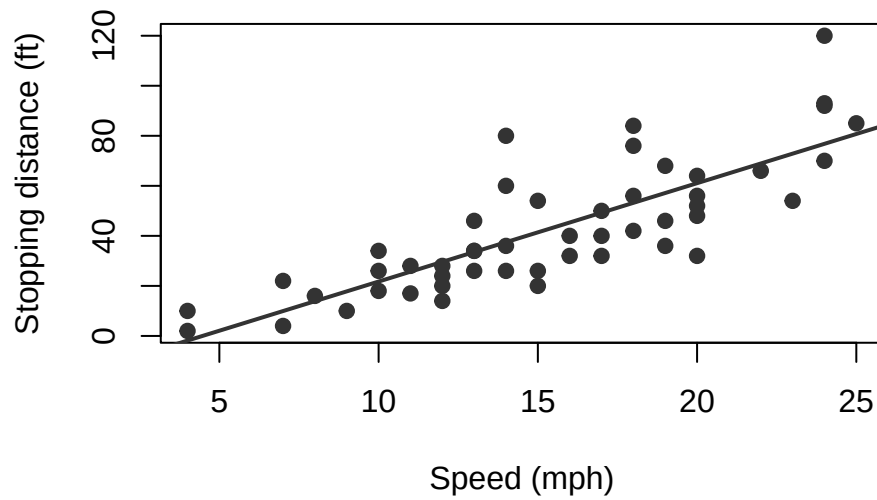


Figure 3.1: Speed and stopping distance

Chapter 4

Discussion and conclusions

4.1 Discussion

Managing the code and manuscript in one project allows tables, figures, and numerical results to be regenerated together. Researchers must separately evaluate the validity of their analyses.

4.2 Conclusions

This example is a starting point for rendering thesis metadata and chapters to PDF. Replace the example title, names, abstracts, data, and manuscript with your own research before submission.

Bibliography

Posit. n.d. “Quarto.” Accessed October 7, 2026. <https://quarto.org/>.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.

Chapter A

Additional analysis

Appendices follow the bibliography and precede the Korean abstract.

Table A.1: Regression coefficients

Term	Estimate
(Intercept)	-17.579
speed	3.932

국문초록

재현 가능한 데이터 분석을 위한 Quarto

기반 학위논문 작성

R 분석 결과와 문서 서식의 통합

홍길동

통계학과

서울대학교 대학원

본 문서는 서울대학교 학위논문 작성을 위한 Quarto 시작 예제이다. 논문 정보, 본문, R 분석 코드를 분리하여 관리하고 하나의 PDF로 출력하는 과정을 보여 준다. 논문 정보는 설정 파일에 입력하며, 장별 내용은 QMD 파일에 작성한다.

예제에서는 R에 포함된 자동차 자료를 사용하여 속도와 제동 거리의 관계를 분석한다. 표, 그림, 수식, 참고문헌과 부록을 포함하여 문서 구성 요소가 함께 출력되는지 확인한다. 예제의 분석 결과는 템플릿 사용법을 설명하기 위한 것이며 실제 학위논문의 연구 결과가 아니다. 제출 전에는 이 초록을 본인의 연구 목적, 방법, 결과 및 결론으로 교체해야 한다.

주요어: 재현성, 데이터 분석, Quarto, R, 학위논문

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